

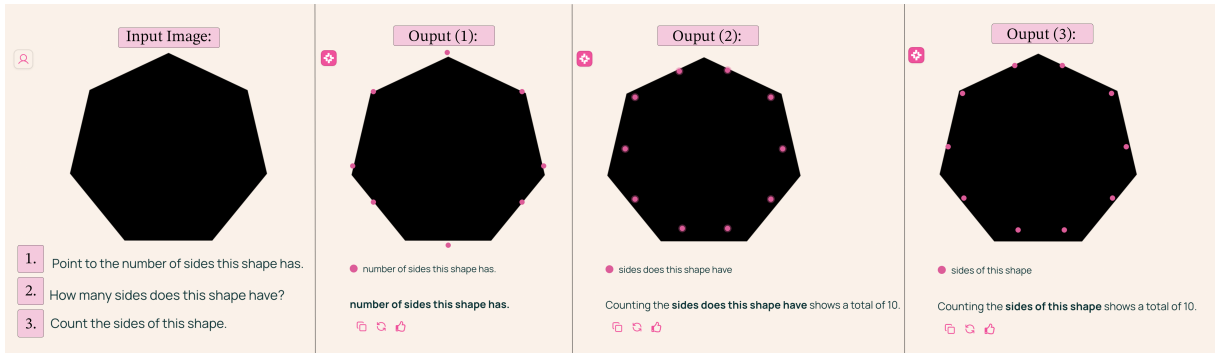


Figure 10: Screenshots from Molmo’s demo website. In all prompts, Molmo misassigns points, counting 10 sides instead of 7.

Image Labels	LLaVA-1.5		LLaVA-Next		LLaVA-OneVision		Qwen2-VL		InternVL		LLaMA-3.2		Math-LLaVA		G-LLaVA		Math-PUMA	
	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT	Direct	VC-CoT
Numbers 1-7	0%	1%	0%	5%	0%	17%	93%	100%	26%	68%	100%	100%	0%	0%	79%	81%	0%	0%
Random Numbers	0%	0%	0%	0%	0%	0%	6%	98%	1%	21%	32%	50%	0%	0%	2%	0%	0%	1%
Letters A-G	0%	0%	0%	0%	0%	0%	32%	92%	1%	2%	45%	96%	0%	0%	0%	0%	0%	0%
Random Letters	0%	0%	0%	0%	0%	0%	0%	83%	0%	3%	3%	58%	0%	0%	0%	0%	0%	1%
	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
No Annotations	0%	0%	0%	0%	0%	0%	12%	99%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%

Table 12: Performance of Visually-Cued Chain-of-Thought (VC-CoT) prompting on heptagons. This table includes models not used in Table 6.

Image Labels	LLaVA-1.5		LLaVA-Next		LLaVA-OneVision		Qwen2-VL		InternVL		LLaMA-3.2		Math-LLaVA		G-LLaVA		Math-PUMA	
	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT	direct	VC-CoT
Numbers 1-15	0%	8%	12%	28%	26%	26%	29%	82%	0%	14%	99%	99%	0%	0%	0%	0%	0%	61%
Random Numbers	0%	0%	0%	0%	0%	0%	0%	5%	0%	3%	32%	21%	0%	0%	0%	0%	0%	28%
Letters A-O	0%	1%	0%	0%	0%	1%	0%	3%	0%	1%	0%	62%	0%	0%	0%	0%	0%	14%
Random Letters	0%	0%	0%	0%	0%	0%	0%	0%	0%	1%	0%	84%	0%	0%	0%	0%	0%	0%
	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
No Annotations	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%

Table 13: Performance of Visually-Cued Chain-of-Thought (VC-CoT) prompting on “arrow on top of plus sign” shape. This table includes models not used in Table 6.